

Incident Response Lifecycle

Incident response is a structured process for detecting, containing, resolving, and learning from operational or security incidents.

1. Preparation

Preparation defines monitoring, alerting, ownership, escalation paths, communication channels, recovery procedures, and access to diagnostic tools.

2. Detection

Detection identifies abnormal behavior using logs, metrics, distributed traces, alerts, health probes, security signals, and user reports.

3. Triage

Triage determines impact, affected users, severity, scope, recent changes, and the teams required to respond.

4. Containment

Containment limits the blast radius and prevents further damage. Examples include disabling a faulty deployment, isolating an affected resource, blocking malicious traffic, or applying temporary resource limits.

5. Eradication

Eradication removes the root cause, malicious artifacts, invalid credentials, faulty configuration, or defective software responsible for the incident.

6. Recovery

Recovery restores normal service safely. Responders validate system health, monitor for recurrence, and return temporarily disabled features in controlled stages.

7. Post-Incident Review

The post-incident review documents impact, timeline, root cause, contributing factors, detection gaps, corrective actions, owners, and target completion dates.

Incident Metrics

Metric	Meaning
Time to Detect	Time between incident start and detection
Time to Acknowledge	Time between alert generation and responder acknowledgement
Time to Mitigate	Time required to reduce customer or system impact

Metric	Meaning
Time to Recover	Time required to restore normal service
Recurrence Rate	Frequency at which similar incidents happen again
Customer Impact	Number of users, tenants, or requests affected

Effective Incident Response Practices

- Assign one incident commander to coordinate the response.
- Maintain a shared timeline of facts, decisions, and actions.
- Separate confirmed facts from hypotheses.
- Use reversible mitigation steps where possible.
- Communicate status at a predictable cadence.
- Track corrective actions until completion.
- Improve alerts, dashboards, and runbooks after each incident.